Modern, Honest, and Open Multimodal Prediction of Post-Earnings-Call Volatility

(working title — living draft, framing-neutral)

Anonymous Author(s)
path-agnostic placeholder

Draft compiled June 24, 2026

Abstract

A decade of work predicts post-earnings-call stock volatility from the text and audio of quarterly conference calls, yet much of this literature compares against weak baselines and may reward models that memorise *which company* is speaking rather than *what* is said. We revisit the task with modern open-weight models, strong econometric baselines, leakage-proof evaluation, and an explicit identity-control suite. We build ECVOL, an open multimodal volatility benchmark over the FinCall-Surprise and MAEC corpora, providing three target families (level log-realised-volatility, change Δv, and HAR-residual), temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 data manifests, and a reproducible evaluation harness. Against persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect baselines, we evaluate a ladder of frozen-representation content models (financial sentence embeddings, FinBERT, self-supervised and prosodic audio features) with light heads, under significance testing (Diebold–Mariano, cluster bootstrap, Holm correction) and identity controls (linear identity probes and same-ticker transcript shuffles). **[TODO:** Finalise headline numbers and the one-sentence verdict once Phases 4–7 land; current evidence is reported framing-neutral in §6.**]** We release all code, configs, manifests, and result tables.

1 Introduction

Earnings conference calls are one of the few recurring events at which management speaks at length, semi-spontaneously, about a public company—in both *what* is said (the transcript) and *how* it is said (the audio). A line of research beginning with Qin and Yang (2019) and continuing through HTML (Yang et al., 2020), MAEC (Li et al., 2020), KeFVP (Niu et al., 2023), and ECC Analyzer (Cao et al., 2024a) claims that multimodal models over this data predict post-call stock volatility better than classical methods.

Two recent observations cast doubt on how much of that progress is real. First, scs (2024) ("Same Company, Same Signal") show that transcript representations used across this literature predominantly encode *company identity* rather than content: same-ticker transcripts cluster tightly, and training-free baselines built on a company's own past volatility beat all transcript-based models on their evaluation. Because volatility is strongly autocorrelated and company-specific, a model that learns "which company is this" inherits most of the apparent skill. Second, a 2025 survey of financial NLP (fin, 2025) finds that only $\sim$14% of reported results reproduce exactly, attributing the gap to dead data links, API drift, and contamination. Together these suggest that the field's leaderboard may be partly an artifact of identity leakage, weak baselines, and irreproducibility.

This paper reworks an abandoned 2024 study in this lineage from the ground up, with three goals. (i) **Modernise:** replace 2019-era components (GloVe, Praat-only acoustics, hand-rolled hierarchical transformers) with current open-weight models—financial sentence embeddings, FinBERT, self-supervised speech and prosodic features, and instruction-tuned LLMs with constrained decoding. (ii) **Open & reproduce:** use only openly licensed data and open-weight models, pin the environment, and release everything needed to rebuild the study, designing explicitly against the reproducibility crisis. (iii) **Evaluate honestly:** measure every model against strong econometric baselines under identity-aware controls and leakage-proof splits, so that any positive claim is attributable to content rather than identity.

Concretely, we ask four research questions:

RQ1 Does earnings-call *content* (text and/or audio) add predictive signal for post-call volatility beyond past-volatility persistence (HAR-RV, GARCH)?

RQ2 Is multimodal (audio + text) better than the best unimodal model, after controlling for past volatility?

RQ3 Do LLM-extracted *structured, auditable* features (guidance, hedging, evasiveness, . . .) outperform opaque dense embeddings?

RQ4 Do any positive results survive (a) ticker-identity controls, (b) ticker-disjoint splits, and (c) evaluation on post-LLM-knowledge-cutoff calls?

The study is pre-registered so that no single outcome strands the work: a clean negative or mixed answer to RQ1/RQ4 is itself a publishable contribution. The paper's framing is decided by the data at a pre-registered gate (§7), not retro-fitted to it; this draft is written *framing-neutral* while the audio, fusion, LLM, and lookahead experiments are completed.

Contributions.

1. **ECVOL, an open multimodal volatility benchmark** over FinCall-Surprise (primary) and MAEC (secondary): three target families (level v, change Δv, HAR-residual), temporal and ticker-disjoint splits with a 30-trading-day embargo, SHA-256 data manifests, and a reproducible harness.
2. **Honest baselines everywhere**: persistence, EWMA, HAR-RV, GARCH(1,1), and ticker-fixed-effect models reported for every horizon, split, and target variant—the comparison most prior work omits.
3. **A modern open-model feature ladder on consumer hardware**: financial embeddings, FinBERT, WavLM / emotion2vec+ / eGeMAPS audio features, fusion heads, and LLM-extracted structured features—all runnable on a single 16–24 GB GPU.
4. **A diagnostic suite** for this task family: identity probes, transcript-shuffle controls, ticker-disjoint evaluation, lookahead-bias tests on post-cutoff data, and a gender-confound analysis.

2 Related Work

The benchmark lineage. Qin and Yang (2019) introduced text+audio volatility prediction and the 572-call EarningsCall dataset, establishing the 3/7/15/30-trading-day log-realised-volatility target family that the field still uses. MAEC (Li et al., 2020) scaled this to ∼3,443 sentence-aligned multimodal calls over the S&P 1500. HTML (Yang et al., 2020) introduced a hierarchical transformer with multi-task learning across horizons; VolTAGE (Sawhney et al., 2020) added graph convolution over inter-stock relations with text/audio fusion; DialogueGAT (Sang and Bao, 2022) modelled speaker/dialogue structure with graph attention. KeFVP (Niu et al., 2023) combined knowledge-enhanced pre-training with a price-conditioned time-series module and is the standard SOTA reference. More recent work moves toward LLMs: ECC Analyzer (Cao et al., 2024a) uses hierarchical LLM summarisation with a RAG "question bank" and wav2vec2 features; RiskLabs (Cao et al., 2024b) fuses calls with news and market series; AT-FinGPT (atf, 2025) is an audio-text LLM. Numeral-aware variants (NumHTML (Yang et al., 2022), NAM/ECNum (Chen et al., 2021a), GNAVol (Shi et al., 2023)) and Q&A-structure models (Ye et al., 2020) target specific aspects of call language.

Audio and prosody. Chen et al. (2024) report that physics-informed acoustic features explain up to 43.8% of 30-day realised-volatility variance, motivating an audio ladder; the FinAudio benchmark (Cao et al., 2025) cautions that audio-LLMs struggle on long financial audio, informing our preference for frozen pooled features over end-to-end audio-LLMs in v1.

Known pitfalls (the methodological core). Our design is organised around four documented failure modes. *(1) Ticker-identity leakage* (scs, 2024) is the central threat to validity (§1); we respond with identity-robust targets (Δv, HAR-residual), a ticker-fixed-effect baseline, and an identity-control suite (§5). *(2) LLM lookahead bias* (Gao et al., 2024; dat, 2025): a 2024-cutoff model "predicting" 2019–2021 outcomes it has effectively seen; we respond with a planned post-cutoff test set and a prompt-masking ablation. *(3) Weak baselines*: much prior work omits persistence, GARCH(1,1), and HAR-RV (Corsi, 2009), which are hard to beat at long horizons; we compute these first and use persistence as the denominator of our headline metric (Gu et al., 2020). *(4) Gender bias in audio features* (Wang et al., 2024; emo, 2024; Mathur et al., 2022): we analyse and report a gender confound rather than debias. We deliberately exclude agentic trading systems and time-series foundation models from v1 scope.

3 Data

3.1 Corpora

We build on two openly licensed multimodal earnings-call corpora and reserve a third, fresh corpus for a lookahead test.

FinCall-Surprise (fin, 2024) (primary; Apache-2.0) contains 2,688 calls spanning 2019–2021 with full MP3 audio, JSON transcripts, and slides. **MAEC** (Li et al., 2020) (secondary; CC-BY-SA-4.0) contains ∼3,443 calls over 2015–2018 (S&P 1500) with sentence-aligned audio features and transcripts but no raw audio. A fresh **2025–2026 post-cutoff corpus** ($\geq$200 calls) is acquired by released scripts for the lookahead experiment (§5); raw audio/transcripts from sources with unclear licenses are never redistributed (a *scripts-not-data* policy).

FinCall-Surprise ships without ticker/date metadata, which we reconstruct from transcript content and cross-check against price data; identity reconstruction resolves 2,496/2,688 earnings calls (92.9%) and passes a 60/60 manual audit.

3.2 Price data and targets

Daily adjusted-close OHLCV is obtained from a free, no-account source and cross-checked against Tiingo on a random ticker sample (adjusted-return correlation > 0.999); delisted or unmatched tickers produce an explicit coverage report rather than being silently dropped. After price joining, FinCall-Surprise yields 2,424 calls with at least one target (97.1% join rate) and MAEC yields 2,578 (75.4%; a documented shortfall driven by 2015–2018 delistings), for a combined modelling set of 5,002 calls.

For each call we define returns $r_t = (P_t - P_{t-1})/P_{t-1}$ relative to day $t = 0$, the last trading day before the call's information is public. Because <4% of calls carry usable timestamps, we apply an after-hours rule (calls assumed to occur after market close; t=1 is the next trading session) and flag it. The primary target is log realised volatility over horizon $\tau \in \{3, 7, 15, 30\}$ trading days,

$$v_{\text{post}}(\tau) = \ln\left(\sqrt{\tfrac{1}{\tau}\textstyle\sum_{t=1}^{\tau}(r_t - \bar{r})^2}\right), \qquad (1)$$

with pre-call volatility $v_{\text{pre}}(\tau)$ defined over the τ sessions ending at day 0. We add two identity-robust variants: the **change** $\Delta v(\tau) = v_{\text{post}}(\tau) - v_{\text{pre}}(\tau)$, which subtracts the company's own level, and the **HAR-residual** $v_{\text{post}}(\tau) - \widehat{\text{HAR}}(\tau)$, deviation from a train-only HAR-RV (Corsi, 2009) forecast. Insufficient post-call history and zero-variance windows yield `NaN` targets excluded with a reason code.

3.3 Leakage-proof splits

Three split families are generated once as committed CSVs and guarded by leakage assertions in CI. *Temporal*: train < validation < test in strict calendar order (∼70/10/20) with a **30-trading-day embargo** between segments so that no 30-day target window crosses a boundary—a constraint most prior work ignores. *Ticker-disjoint*: companies in validation/test never appear in training (grouped, seeded), the headline identity-control split. *Combined*: both constraints simultaneously, reported as the hardest robustness row. An information rule is codified and enforced: every feature function receives an `as_of` timestamp, and the harness fails any feature that reads price data after it. FinCall-Surprise spans COVID; per-year breakdowns are mandatory and a Feb–Apr 2020 exclusion is reported as an exploratory sensitivity.

4 Methods

We organise models as a ladder (Stages 0–5; Figure 1). Two invariants make the comparison honest. First, from Stage 2 upward every learned model includes past-volatility covariates (v_{pre} at all horizons plus the HAR-RV forecast) and is *also* reported with them ablated, baking the persistence control into every comparison. Second, given the small-data regime (10^3–10^4 samples) we use frozen representations with light heads (ridge, shallow MLP, LightGBM) and per-horizon models by default; fine-tuning is gated to Stage 6 and a shared multi-task head is an ablation, not the headline.

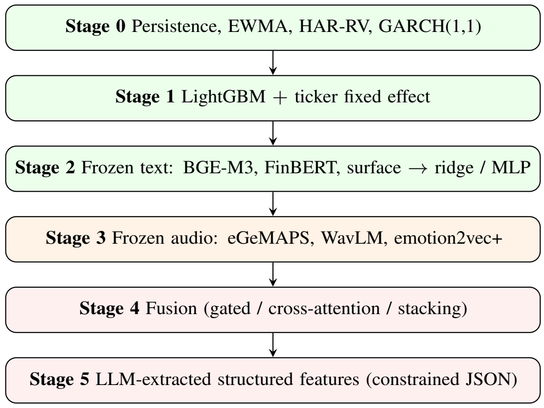

Figure 1: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Green = complete, orange = in progress, red = planned. **[TODO:** update shading as phases land**]**.

Stage 0 — econometric baselines. Persistence ($\hat{v} = v_{\text{pre}}$), EWMA (RiskMetrics λ=0.94), HAR-RV (OLS on daily/weekly/monthly realised-variance factors, Corsi, 2009), and GARCH(1,1) (per-call fit, `arch`). These establish the floor and the denominator of the headline metric.

Stage 1 — tabular identity baseline. LightGBM over past-volatility features, trivial metadata (transcript length, Q&A turn count where available), and a **ticker fixed effect**. This is the in-house "Same-Company-Same-Signal" control: it measures how much apparent multimodal performance is identity plus metadata.

Stage 2 — frozen text content. We segment each transcript into prepared-remarks vs. Q&A and speaker turns, then extract three feature blocks: BGE-M3 (BAAI, 2023) section-pooled sentence embeddings (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates per scope×role, and surface statistics. Heads are ridge (with the α grid selected on validation, *no* retrain on train+val) and a shallow MLP (1×256, L2 + early stopping, 5 seeds). Features use fp32 with content-hash caching for bit-identical re-extraction.

Stage 3 — frozen audio content. Audio is QC'd and resampled to 16 kHz mono FLAC. We extract eGeMAPSv02 functionals (88-d, openSMILE (Eyben et al., 2010), CPU, interpretable), then WavLM-Large (Chen et al., 2021b) and emotion2vec+ (Ma et al., 2023) chunk embeddings mean-pooled to a per-call vector, fed to the same heads. We use per-call pooling only (no per-sentence alignment), and report a gender-confound analysis. *[pending: WavLM and emotion2vec+ full-corpus extraction + heads (T4.3–T4.4).]*

Stage 4 — fusion. Gated-fusion and cross-attention heads over frozen modality embeddings, plus late-fusion stacking with the Stage-1 GBDT, to test whether multimodal beats the best unimodal model (RQ2). *[pending: Phase 5.]*

Stage 5 — LLM-extracted structured features. Qwen2.5-7B-Instruct (Qwen Team, 2024) (4-bit) with Outlines-constrained JSON (dottxt-ai, 2023) extracts auditable per-section features (guidance direction, hedging intensity, Q&A evasiveness, surprise mentions, analyst tone) fed into the Stage-1 GBDT. A human-audit gate ($\kappa > 0.6$ on 50 calls) precedes corpus-scale extraction (RQ3). *[pending: Phase 6.]*

5 Experimental Protocol

Metrics. We report MSE on log-RV per horizon (literature comparability); R^2_{OOS} vs. persistence (Gu et al., 2020; $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$), the headline metric, which is not gameable by identity memorisation the way raw MSE is; MAE; and cross-sectional Spearman rank correlation per calendar quarter. All metrics are reported on all three targets (v, Δv, HAR-residual) $\times$ all splits.

Statistical rigor. Every learned model is run with $\geq$5 seeds (mean $\pm$ std). We use Diebold–Mariano tests (Diebold and Mariano, 1995) (HLN small-sample correction) on per-call squared-error differentials vs. persistence and vs. the best preceding stage; cluster-bootstrap confidence intervals (1,000 resamples) clustered by ticker and by calendar quarter; and Holm-corrected p-values across horizons for confirmatory claims. Confirmatory vs. exploratory analyses are labelled in advance: confirmatory analyses are Stage-k vs. Stage-1 DM tests on Δv at each τ, Stage 4 vs. best unimodal, Stage 5 vs. Stage 2, and the temporal-vs-ticker-disjoint gap per stage; all per-year/regime, COVID-exclusion, prepared-vs-Q&A, call-length, and gender-confound analyses are exploratory.

Identity-control suite. The signature diagnostic of this study probes whether content models read content or identity, via four controls. *(1) Ticker-only model*: predict the per-ticker target mean (global mean for unseen tickers); if a content model matches it on level-v, that model reads identity. *(2) Same-ticker transcript shuffle*: replace each test call's transcript with a different call from the *same* ticker; if performance is unchanged the model reads identity, not content, whereas a *global* shuffle (any other call) isolates pure overfitting. *(3) Identity probe*: a linear probe predicting the ticker from frozen call embeddings; high accuracy demonstrates identity is encoded. *(4) Ticker-disjoint split* results reported beside temporal-split results in every table.

Lookahead-bias experiments. *[pending: Phase 7.]* The entire frozen pipeline—no retraining, no post-hoc threshold changes (rule fixed in advance)—will be evaluated on the 2025–2026 post-cutoff corpus; disproportionate degradation of Stage-5 LLM features relative to other stages would be contamination evidence. A masking ablation (company names, tickers, dates removed from Stage-5 prompts) will estimate identity-mediated leakage.

Reproducibility. All results regenerate byte-identically from committed CSV artifacts via `ecvol report`, guarded by a CI test. Every data file carries a SHA-256 manifest entry, and every run artifact is keyed by config hash plus git SHA.

6 Results

We report results factually and defer the headline framing to §7. Curated headline tables appear here on the identity-robust Δv target for FinCall-Surprise; the complete grid (both corpora, level-v and Δv, R^2_{OOS} and MSE, all splits and horizons) is in Appendix A. Stars denote Diebold–Mariano significance ($p < 0.05$): in Table 1 vs. persistence, and in Table 2 vs. the Stage-1 ticker-FE GBDT.

6.1 Baselines (Stages 0–1)

Table 1 shows the econometric floor. HAR-RV is strong at short and medium horizons on both splits, confirming RQ1's premise that past-volatility baselines are hard to beat. The one striking failure is the FinCall *temporal* split at τ=30, where every model falls below persistence (HAR-RV $R^2_{\text{OOS}} = -0.287$): the training window is COVID-inflated (Feb–May 2020) while the test window is calm late 2021, so a long-horizon level forecast trained on the crisis regime is mis-calibrated. The same models pass comfortably on the ticker-disjoint split (HAR-RV $R^2_{\text{OOS}} = +0.229$ at τ=30) and on MAEC, so we treat this as a documented regime exception rather than a target bug; per-year breakdowns are mandatory in the appendix.

6.2 Text content (Stage 2)

Table 2 shows frozen-text heads against the Stage-1 baseline on Δv. The result is a clean negative: *text content does not cleanly beat the identity/past-volatility baselines*. On the FinCall temporal split the text ridge head is near zero and collapses at τ=30; the MLP heads overfit badly. On the ticker-disjoint split the text ridge head is weak (R^2_{OOS} between -0.04 and

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=481)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108

Table 1: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. $*$: Diebold–Mariano significant vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=481)				
Ridge (text)	0.028*	0.030*	0.015	-0.144*
Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
Ticker-disjoint split (n=484)				
Ridge (text)	-0.044*	-0.026*	0.008	0.171
Ridge (past-vol)	0.451	0.306	0.191*	0.224*
Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (past-vol)	0.492*	0.363*	0.230*	0.249*

Table 2: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head; adding text to past-vol does not help. $*$: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

+0.17) while the past-vol-only head is consistently strong ($R^2_{\text{OOS}} \approx 0.19$–$0.45$), so any apparent text gain is not separable from what past volatility already provides. MAEC shows the same pattern at smaller magnitude (Appendix A).

6.3 Identity controls (Stage 2 diagnostic)

Table 3 explains the negative result. A linear probe recovers the ticker from frozen call embeddings at 89.5% on FinCall (319× chance) and 53.9% on MAEC (487× chance): identity is heavily encoded. The transcript shuffle is decisive—replacing each test call's transcript with a *different call from the same ticker* barely changes R^2_{OOS}, whereas a global shuffle degrades it. The heads are reading *which company* is speaking, not *what* is said. This is the in-corpus confirmation of scs (2024) and the reason RQ4 is central.

	Identity probe (linear)	
Corpus	Accuracy	× chance
FinCall (n=2656, 357 tickers)	89.5%	319×
MAEC (n=3132, 902 tickers)	53.9%	487×

Condition	τ=3	τ=7	τ=15	τ=30
Real transcript	0.028	0.030	0.015	-0.144
Within-ticker shuffle	0.017	0.020	0.006	-0.142
Global shuffle	-0.099	-0.121	-0.115	-0.266

Table 3: Identity controls. Top: a linear probe predicts the ticker from frozen BGE-M3 call embeddings far above chance, demonstrating identity is heavily encoded. Bottom: R^2_{OOS} of the text ridge head on the Δv target (FinCall, temporal split, test, n=481). Replacing each test call's transcript with a *different call from the same ticker* (within-ticker shuffle) barely changes performance, whereas a global shuffle degrades it—evidence the head reads ticker identity rather than call content.

6.4 Audio, fusion, LLM, and lookahead (pending)

[pending: Stage 3 audio (Result Table 3): WavLM-Large and emotion2vec+ full-corpus extraction and heads; eGeMAPS (88-d) already extracted for all 2,671 calls.] [pending: Stage 4 fusion (Result Table 4): the main §5 ablation grid.] [pending: Stage 5 LLM structured features (Result Table 5) and the masking ablation.] [pending: Phase 7 post-cutoff lookahead evaluation on 2025–2026 calls.] Each lands as its phase completes; the framing gate (§7) is re-evaluated when the audio results are in.

7 Discussion

What the evidence so far shows. On finished phases, three statements hold. (i) Strong econometric baselines are hard to beat, and the headline metric (R^2_{OOS} vs. persistence) exposes a long-horizon regime failure (FinCall temporal τ=30) that raw MSE would obscure. (ii) Frozen text content does not cleanly add signal over past-volatility and identity baselines on the identity-robust Δv target. (iii) The reason is measurable: call embeddings encode ticker identity (89.5% probe accuracy on FinCall), and within-ticker transcript shuffles leave performance essentially unchanged—an in-corpus replication of the "Same Company, Same Signal" effect (scs, 2024).

The framing gate (pre-registered, still open). The paper's spine is decided at a pre-registered gate, not retro-fitted. *Path A (positive showcase)* is adopted only if content-bearing models beat the Stage-0/1 floor on Δv or HAR-residual with DM-significant improvements that survive the identity-control suite on at least two of four horizons. *Path B (rigorous re-examination)* is adopted otherwise. The text-only evidence currently points toward Path B, but the gate is not yet final: it is re-evaluated once the audio ladder (Stage 3) completes, since prosody is the modality most plausibly carrying content orthogonal to identity (Chen et al., 2024), and again after fusion (Stage 4), LLM features (Stage 5), and the post-cutoff lookahead test (Phase 7). This draft is therefore written framing-neutral. **[TODO:** Resolve the gate and rewrite the abstract/intro headline once Stage 3 lands; record the decision in the project DECISION log.**]**

Why the contribution holds either way. Independent of the gate, ECVOL contributes an open, identity-controlled, leakage-proof multimodal benchmark with honest baselines and a reusable diagnostic suite—precisely the apparatus the field has lacked. A negative or mixed answer to RQ1/RQ4 is a publishable result that the existing literature does not provide.

8 Conclusion

We revisit multimodal earnings-call volatility prediction with modern open-weight models, strong econometric baselines, leakage-proof splits, and an explicit identity-control suite, released as the ECVOL benchmark. On the completed text ladder, frozen content does not cleanly beat past-volatility and identity baselines, and identity probes plus transcript-shuffle controls show that call embeddings encode *which company* is speaking more than *what* is said. The audio, fusion, LLM, and post-cutoff-lookahead experiments are in progress and will resolve the pre-registered framing gate; this draft reports the current evidence framing-neutral. **[TODO:** Finalise the conclusion's verdict after the framing gate is resolved.**]**

Limitations

Scope of current evidence. The negative finding is established on the text ladder only; audio (Stage 3), fusion (Stage 4), and LLM-structured features (Stage 5) are not yet complete, so claims about *multimodal* content are deferred. **Timestamps.** Fewer than 4% of calls carry usable timestamps, so we apply an after-hours rule; mis-dated calls would add noise to short-horizon targets. **Regime sensitivity.** FinCall-Surprise spans COVID; the temporal τ=30 baseline failure shows long-horizon level forecasts are regime-sensitive, which is why we emphasise the identity-robust Δv target and report per-year breakdowns. **MAEC coverage.** The 75.4% price-join rate (2015–2018 delistings) makes MAEC a secondary, comparability corpus. **Lookahead.** LLM-feature results on 2019–2021 calls are suspect by default until the Phase 7 post-cutoff evaluation and masking ablation quantify contamination. **No diarisation in v1.** Audio uses per-call pooling, not speaker-resolved features. **[TODO:** Revisit and tighten as later phases land.**]**

Ethics Statement

Data and licensing. We use only openly licensed corpora (FinCall-Surprise, Apache-2.0; MAEC, CC-BY-SA-4.0) and open-weight models. Derived numeric features inherit and explicitly state their upstream license; raw audio/transcripts from unclear-license sources are never redistributed (*scripts-not-data*). **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora, which can produce error disparities (Wang et al., 2024; emo, 2024; Mathur et al., 2022); we analyse and report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice; predictions are noisy and regime-sensitive, and should not be used for automated trading. **Reproducibility as an ethical commitment.** Given the documented reproducibility crisis in financial NLP (fin, 2025), we pin the environment, checksum every data file, and regenerate all result tables byte-identically from committed artifacts.

References

Emo-bias: A large scale evaluation of social bias on speech emotion recognition. arXiv:2406.05065, 2024. URL https://arxiv.org/abs/2406.05065.

FinCall-Surprise: An earnings-call dataset with audio, transcripts, and slides. arXiv:2510.03965; data: github.com/Tizzzzy/FinCall-Surprise (Apache-2.0), 2024. URL https://arxiv.org/abs/2510.03965.

Same company, same signal: The role of identity in earnings call transcripts. arXiv:2412.18029, 2024. URL https://arxiv.org/abs/2412.18029.

AT-FinGPT: Financial risk prediction via an audio-text large language model. *Finance Research Letters*, 2025. URL https://www.sciencedirect.com/science/article/abs/pii/S1544612325002314.

DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838, 2025. URL https://arxiv.org/abs/2603.11838.

Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274, 2025. URL https://arxiv.org/abs/2504.07274.

BAAI. BGE sentence embeddings. Hugging Face (MIT), 2023. URL https://huggingface.co/BAAI/bge-large-en-v1.5.

Yupeng Cao et al. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*, 2024a. URL https://arxiv.org/abs/2404.18470.

Yupeng Cao et al. RiskLabs: Predicting financial risk using large language model based on multimodal and multi-sources data. arXiv:2404.07452, 2024b. URL https://arxiv.org/abs/2404.07452.

Yupeng Cao et al. FinAudio: A benchmark for audio large language models in financial applications. arXiv:2503.20990, 2025. URL https://arxiv.org/abs/2503.20990.

Chen et al. Distilling numeral information for volatility forecasting. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, 2021a. doi: 10.1145/3459637.3482089.

Chen et al. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility. arXiv:2508.18653, 2024. URL https://arxiv.org/abs/2508.18653.

Sanyuan Chen et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large, 2021b. URL https://huggingface.co/microsoft/wavlm-large.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 2009. doi: 10.1093/jjfinec/nbp001.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 1995. doi: 10.1080/07350015.1995.10524599.

dottxt-ai. Outlines: Structured generation / constrained decoding. github.com/dottxt-ai/outlines, 2023. URL https://github.com/dottxt-ai/outlines.

Florian Eyben, Martin Wöllmer, and Björn Schuller. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python, 2010. URL https://github.com/audeering/opensmile-python.

Gao, Jiang, and Yan. A test of lookahead bias in LLM forecasts. arXiv:2512.23847, 2024. URL https://arxiv.org/abs/2512.23847.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 2020. doi: 10.1093/rfs/hhaa009.

Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020. doi: 10.1145/3340531.3412879.

Ziyang Ma et al. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large, 2023. URL https://huggingface.co/emotion2vec/emotion2vec_plus_large.

Puneet Mathur et al. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.139/.

Hao Niu et al. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. URL https://aclanthology.org/2023.findings-emnlp.770/.

ProsusAI. FinBERT: Financial sentiment analysis with bert. Hugging Face, 2019. URL https://huggingface.co/ProsusAI/finbert.

Yu Qin and Yi Yang. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019. URL https://aclanthology.org/P19-1038/.

Qwen Team. Qwen2.5-7B-Instruct. Hugging Face (Apache-2.0), 2024. URL https://huggingface.co/Qwen/Qwen2.5-7B-Instruct.

Yunxin Sang and Yang Bao. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL https://aclanthology.org/2022.findings-emnlp.117/.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. VolTAGE: Volatility forecasting via text-audio fusion with graph convolution networks. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. URL https://aclanthology.org/2020.emnlp-main.643/.

Shi et al. Enhancing volatility forecasting in financial markets: A general numeral attachment dataset for understanding earnings calls.

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.354*	0.262*	-0.040	-0.839*

Table 4: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=481).

In *Proceedings of IJCNLP-AACL 2023*, 2023. URL https://aclanthology.org/2023.ijcnlp-short.5/.

Wang et al. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324, 2024. URL https://arxiv.org/abs/2407.18324.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*, 2020.

Linyi Yang et al. NumHTML: Numeric-oriented hierarchical transformer model for multi-task financial forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. URL https://arxiv.org/abs/2201.01770.

Ye, Qin, and Xu. Financial risk prediction with multi-round q&a attention network. In *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, 2020. URL https://www.ijcai.org/proceedings/2020/631.

A Complete Results

The tables below are generated byte-identically by `ecvol report` from the committed result CSVs (`data/results/`) and are the source of truth for the curated tables in §6; the curated tables copy selected cells from them. They span both corpora (FinCall-Surprise, MAEC), the level-v and Δv targets, both the R^2_{OOS} and MSE metrics, and the temporal and ticker-disjoint splits at all horizons.

A.1 Stage-0/1 baselines (Result Table 1)

A.2 Stage-2 text heads (Result Table 2)

[TODO: Add Result Tables 3 (audio), 4 (fusion), 5 (LLM) here as phases complete; they will be emitted by `ecvol report` in the same format.]

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.412*	0.278*	0.078	0.116*

Table 5: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.387*	0.362*	0.145*	-0.219*
HAR-RV	0.464*	0.410*	0.207*	-0.287*
GARCH(1,1)	0.265*	0.193*	-0.140*	-0.541*
GBDT (ticker FE)	0.344*	0.248*	-0.057	-0.904*

Table 6: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.357*	0.199*	-0.009	0.046
HAR-RV	0.495*	0.357*	0.211*	0.229*
GARCH(1,1)	0.356*	0.229*	0.088	0.209*
GBDT (ticker FE)	0.407*	0.275*	0.089	0.108

Table 7: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.444*	0.301*	0.120*	-0.144*
HAR-RV	0.507*	0.392*	0.306*	0.206*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*
GBDT (ticker FE)	0.397*	0.333*	0.217*	0.131*

Table 8: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.398*	0.297*	0.119*	-0.144*
HAR-RV	0.474*	0.379*	0.311*	0.225*
GARCH(1,1)	0.306*	0.213*	0.009	-0.321*
GBDT (ticker FE)	0.395*	0.306*	0.236*	0.221*

Table 9: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.444*	0.301*	0.120*	-0.144*
HAR-RV	0.507*	0.392*	0.306*	0.206*
GARCH(1,1)	0.387*	0.281*	0.054	-0.309*
GBDT (ticker FE)	0.407*	0.340*	0.213*	0.186*

Table 10: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
EWMA	0.398*	0.297*	0.119*	-0.144*
HAR-RV	0.474*	0.379*	0.311*	0.225*
GARCH(1,1)	0.306*	0.213*	0.009	-0.321*
GBDT (ticker FE)	0.394*	0.285*	0.227*	0.233*

Table 11: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.174	0.419	0.208	0.111
EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.758*	0.309*	0.216	0.205*

Table 12: fincall — temporal split — level-v — MSE (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.142	0.386	0.214	0.233
EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.672*	0.278*	0.197	0.206*

Table 13: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.174	0.419	0.208	0.111
EWMA	0.720*	0.267*	0.178*	0.136*
HAR-RV	0.630*	0.247*	0.165*	0.143*
GARCH(1,1)	0.860*	0.335*	0.233*	0.167*
GBDT (ticker FE)	0.771*	0.315*	0.220	0.212*

Table 14: fincall — temporal split — Δv — MSE (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.142	0.386	0.214	0.233
EWMA	0.735*	0.309*	0.216	0.222
HAR-RV	0.577*	0.248*	0.169*	0.180*
GARCH(1,1)	0.735*	0.297*	0.195	0.184*
GBDT (ticker FE)	0.678*	0.279*	0.195	0.208

Table 15: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.383	0.571	0.318	0.189
EWMA	0.768*	0.399*	0.280*	0.216*
HAR-RV	0.682*	0.347*	0.221*	0.150*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*
GBDT (ticker FE)	0.834*	0.381*	0.249*	0.164*

Table 16: maec — temporal split — level-v — MSE (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.268	0.492	0.302	0.184
EWMA	0.764*	0.346*	0.266*	0.210*
HAR-RV	0.667*	0.306*	0.208*	0.142*
GARCH(1,1)	0.881*	0.384*	0.296	0.242*
GBDT (ticker FE)	0.767*	0.342*	0.230*	0.143*

Table 17: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.383	0.571	0.318	0.189
EWMA	0.768*	0.399*	0.280*	0.216*
HAR-RV	0.682*	0.347*	0.221*	0.150*
GARCH(1,1)	0.848*	0.411*	0.301	0.247*
GBDT (ticker FE)	0.820*	0.377*	0.250*	0.154*

Table 18: maec — temporal split — Δv — MSE (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.268	0.492	0.302	0.184
EWMA	0.764*	0.346*	0.266*	0.210*
HAR-RV	0.667*	0.306*	0.208*	0.142*
GARCH(1,1)	0.881*	0.384*	0.296	0.242*
GBDT (ticker FE)	0.769*	0.352*	0.233*	0.141*

Table 19: maec — ticker_disjoint split — Δv — MSE (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.439*	0.319	0.037	-0.577
MLP (text)	0.147*	-0.241*	-1.101*	-2.633*
Ridge (past-vol)	0.415*	-1.052	-2.025	-0.371*
MLP (past-vol)	0.456*	-0.228	-2.706	-2.477
Ridge (text+vol)	0.448*	0.328	0.045	-1.091
MLP (text+vol)	-2.092	-7.189	-18.384	-26.901

Table 20: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.416	0.194*	-0.047*	0.112
MLP (text)	-0.362*	-1.801*	-3.485*	-2.872*
Ridge (past-vol)	0.432	0.306	0.191*	0.224*
MLP (past-vol)	0.482*	0.365*	0.243*	0.212*
Ridge (text+vol)	0.418	0.211	-0.006	0.139
MLP (text+vol)	-0.254*	-1.556*	-3.005*	-2.486*

Table 21: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.028*	0.030*	0.015	-0.144*
MLP (text)	-0.108*	-0.175*	-0.265*	-1.042
Ridge (past-vol)	0.415*	-1.053	-2.858	-0.667
MLP (past-vol)	0.390	-1.017	-2.249	-0.760
Ridge (text+vol)	-1.678	-3.076	-3.586	-0.468
MLP (text+vol)	-1.186	-1.571	-2.112	-1.295

Table 22: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	-0.044*	-0.026*	0.008	0.171
MLP (text)	-0.195*	-0.271*	-0.339*	-0.204*
Ridge (past-vol)	0.451	0.306	0.191*	0.224*
MLP (past-vol)	0.492*	0.363*	0.230*	0.249*
Ridge (text+vol)	0.223*	0.129*	0.094	0.236
MLP (text+vol)	0.275*	0.140*	-0.061*	0.051

Table 23: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.478*	0.225*	-0.033*	-0.322*
MLP (text)	-0.748*	-1.753*	-3.272*	-6.016*
Ridge (past-vol)	0.437	0.222*	0.168	0.208
MLP (past-vol)	0.472*	0.375	0.301*	0.247*
Ridge (text+vol)	0.482*	0.279	0.132	-0.013*
MLP (text+vol)	-0.405*	-1.460*	-2.977*	-5.058*

Table 24: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.466*	0.277	0.117*	-0.036*
MLP (text)	-0.115*	-0.956*	-1.583*	-2.635*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.482*	0.369*	0.281	0.250
Ridge (text+vol)	0.451	0.299	0.186	0.090*
MLP (text+vol)	-0.077*	-0.884*	-1.418*	-2.992*

Table 25: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.017*	0.013*	0.047*	-0.074*
MLP (text)	-0.301*	-0.164*	-0.338*	-0.511*
Ridge (past-vol)	0.455	0.313	0.244	0.212
MLP (past-vol)	0.497*	0.395*	0.323*	0.261*
Ridge (text+vol)	-0.005*	-0.142*	0.211	0.138
MLP (text+vol)	0.259*	0.242*	0.068*	-0.080*

Table 26: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.049*	0.056*	0.048*	0.047*
MLP (text)	-0.051*	-0.108*	-0.165*	-0.199*
Ridge (past-vol)	0.455*	0.329	0.261	0.262
MLP (past-vol)	0.489*	0.389*	0.328*	0.294*
Ridge (text+vol)	0.330	0.142*	0.204	0.163
MLP (text+vol)	0.366	0.244	0.120*	-0.111*

Table 27: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.659*	0.285	0.200	0.176
MLP (text)	1.001*	0.519*	0.437*	0.405*
Ridge (past-vol)	0.687*	0.859	0.629	0.153*
MLP (past-vol)	0.639*	0.514	0.771	0.387
Ridge (text+vol)	0.648*	0.281	0.199	0.233
MLP (text+vol)	3.631	3.428	4.032	3.109

Table 28: fincall — temporal split — level-v — MSE (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.667	0.311*	0.224*	0.207
MLP (text)	1.556*	1.080*	0.958*	0.903*
Ridge (past-vol)	0.649	0.268	0.173*	0.181*
MLP (past-vol)	0.592*	0.245*	0.162*	0.184*
Ridge (text+vol)	0.665	0.304	0.215	0.201
MLP (text+vol)	1.432*	0.986*	0.856*	0.813*

Table 29: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.141*	0.406*	0.205	0.127*
MLP (text)	1.302*	0.492*	0.263*	0.228
Ridge (past-vol)	0.687*	0.860	0.803	0.186
MLP (past-vol)	0.717	0.844	0.676	0.196
Ridge (text+vol)	3.145	1.706	0.954	0.164
MLP (text+vol)	2.567	1.076	0.647	0.256

Table 30: fincall — temporal split — Δv — MSE (test) (n=481).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.193*	0.396*	0.212	0.193
MLP (text)	1.365*	0.490*	0.286*	0.281*
Ridge (past-vol)	0.627	0.268	0.173*	0.181*
MLP (past-vol)	0.580*	0.246*	0.164*	0.175*
Ridge (text+vol)	0.888*	0.336*	0.194	0.178
MLP (text+vol)	0.828*	0.332*	0.227*	0.221

Table 31: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.722*	0.443*	0.329*	0.250*
MLP (text)	2.416*	1.573*	1.359*	1.325*
Ridge (past-vol)	0.778	0.445*	0.265	0.150
MLP (past-vol)	0.730*	0.357	0.222*	0.142*
Ridge (text+vol)	0.716*	0.412	0.276	0.191*
MLP (text+vol)	1.943*	1.405*	1.266*	1.144*

Table 32: maec — temporal split — level-v — MSE (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.677*	0.356	0.266*	0.190*
MLP (text)	1.413*	0.963*	0.779*	0.668*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.656*	0.311*	0.217	0.138
Ridge (text+vol)	0.695	0.345	0.246	0.167*
MLP (text+vol)	1.365*	0.927*	0.729*	0.734*

Table 33: maec — ticker_disjoint split — level-v — MSE (test) (n=514).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.358*	0.564*	0.303*	0.203*
MLP (text)	1.798*	0.665*	0.426*	0.285*
Ridge (past-vol)	0.753	0.392	0.241	0.149
MLP (past-vol)	0.695*	0.346*	0.215*	0.140*
Ridge (text+vol)	1.390*	0.653*	0.251	0.163
MLP (text+vol)	1.025*	0.433*	0.297*	0.204*

Table 34: maec — temporal split — Δv — MSE (test) (n=507).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.206*	0.464*	0.287*	0.175*
MLP (text)	1.332*	0.546*	0.351*	0.220*
Ridge (past-vol)	0.691*	0.330	0.223	0.136
MLP (past-vol)	0.648*	0.301*	0.203*	0.130*
Ridge (text+vol)	0.850	0.422*	0.240	0.154
MLP (text+vol)	0.804	0.372	0.265*	0.204*

Table 35: maec — ticker_disjoint split — Δv — MSE (test) (n=514).